

COMP1551

Application Development

Lecture 1: Introduction

Konstantin Kapinchev
University of Greenwich

Application Development

Lecture 1: Introduction

In this lecture:

- **Welcome to the Module**
- **Structure and Content of the Module**
- **Assessment**
- **The Application Development Process**
- **Types of Applications**
- **The Programming Language**
- **Syntax of the Language**
- **The IDE**

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Welcome to the Module

- **Welcome to the Application Development module**
- **This module is focused on the design and the development of object-oriented, data-driven and interactive applications**
- **The module is practically oriented, emphasising the utilisation of programming languages and tools, which are widely used in the industry**
- **Those include Visual Studio, C#, UML and SQL**

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Structure and Content of the Module

- **Timetabled Teaching:**
 - **Lectures, 2-hour weekly**
 - **Laboratory Sessions, 1-hour weekly**
- **Independent Learning**
- **Assessment: a practically oriented coursework**

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Structure and Content of the Module *continued*

- **Lecture Topics**

- **Introduction**
- **The C# Programming Language**
- **Object-Oriented Applications**
- **Principles of Object-Oriented Programming**
- **Object-Oriented Techniques in Practice**
- **Designing Object-Oriented Applications**
- **Graphical User Interface**
- **File Input and Output**
- **Database Management Systems**
- **Structured Query Language**

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Assessment

- **Coursework demonstrates the application development process**
- **C# is the programming language of choice**
- **The laboratory sessions will focus on coursework related tasks**
- **Coursework contributes at 100% of the grade**
- **Coursework deadline is 2 April 2024**

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The Application Development Process

- **It is a step-by-step process, can be organised into stages**
- **Work on each stage is usually done by a dedicated team¹**
- **Two types of overlapping might occur²**
 - **The same team working on different stages**
 - **Simultaneous work on different stages**

¹Especially for the development of larger projects

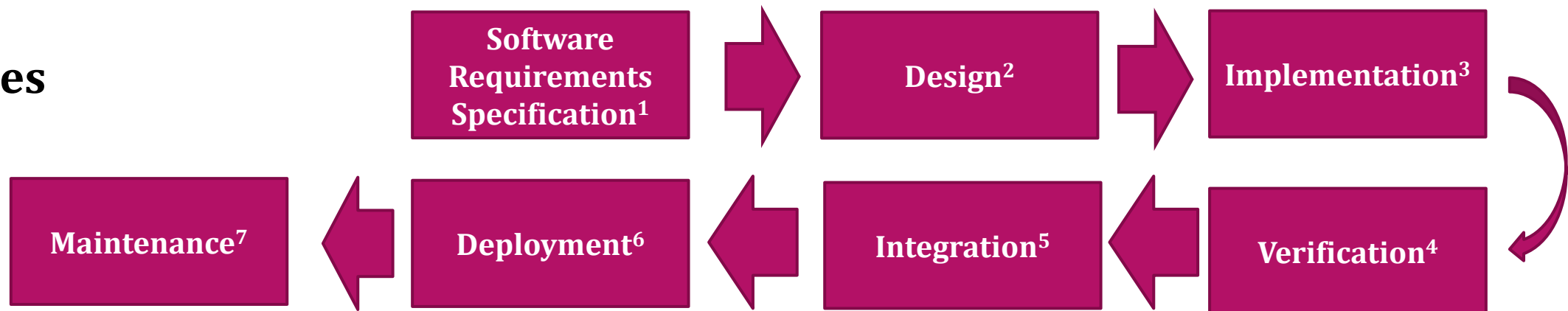
²This might affect negatively the development process

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The Application Development Process *continued*

- Stages



¹Can include functional and non-functional requirements

²A popular tool to design object-oriented applications is UML

³Utilises programming languages, such as C#, SQL and others (work on the codebase)

⁴Involves unit testing, integration testing and system testing

⁵In case of larger projects with multiple components

⁶Installs the software product on the target hardware and software (OS) platforms

⁷Usually involves updates fixing existing bugs or improving performance, security and others

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Types of Applications

- **Examples of some of the more notable applications, classified by their computing platforms:**
 - **Desktop Applications (CLI¹, GUI²)**
 - **Web Applications**
 - **Mobile Apps**
 - **Cloud Computing Applications**
 - **And others ...**

¹Command Line Interface, also called Console Applications

²Graphical User Interface

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The Programming Language

- For this module, the language of choice is C#
- The language is developed by Microsoft in 2000
- It is designed as¹:
 - Imperative
 - Structured
 - Object-oriented

The C# logo, featuring the letters 'C#' in a stylized blue font, is displayed within a light gray square.

¹As a multi-paradigm language, C# includes other paradigms as well

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The Programming Language *continued*

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Note:

Programming Paradigm

Most programming languages have one or more main approaches to solving problems. These main approaches underpin the programs written in those languages.

¹As a multi-paradigm language, C# includes other paradigms as well

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The Programming Language *continued*

The C# logo, featuring the letters 'C#' in a stylized blue font, is positioned on the right side of the slide within a light gray rectangular box.

- The "Hello World" Program
 - Purpose:
 - To test the translator¹
 - To test the output of the environment²
 - To introduce a new programming language to learners
 - To illustrate basic functionality

¹Compiler or interpreter

²For example an IDE such as Visual Studio

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The Programming Language *continued*

- The "Hello World" Program

C#

```
namespace MyNameSpace
{
    class HelloWorld
    {
        static void Main(string[] args)
        {
            System.Console.WriteLine("Hello World");
        }
    }
}
```

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Syntax of the Language

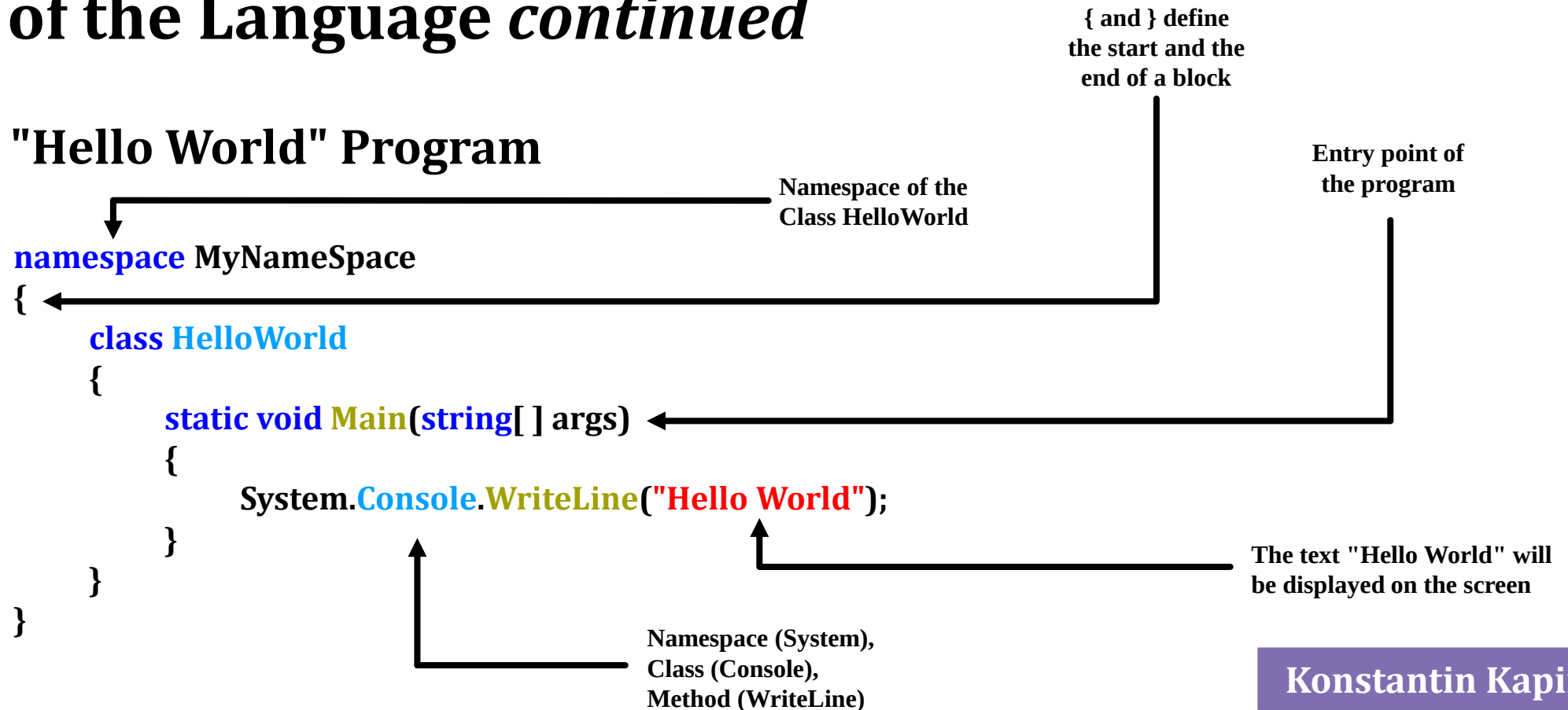
- **The syntax of a programming language defines the rules, which are followed by the language**
- **The syntax determines the correct use of:**
 - **Symbols (tokens, lexemes)**
 - **Structures (operators, statements)**
 - **Context (scope)**
 - **Data types**
 - **And others ...**

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Syntax of the Language *continued*

- The "Hello World" Program



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Syntax of the Language *continued*

- **Namespace**
 - A logically separated section of the source code, classes with similar functionalities usually share the same namespace
- **Block**
 - Groups a number of statements into a single entity, the start of a block is defined with "{", the end of a block is defined with "}"

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Syntax of the Language *continued*

- Entry Point
 - The first method, which is called when a program starts
 - In C# it is called "Main"
 - The following is the signature of the Main method

```
static void Main(string[ ] args)
```

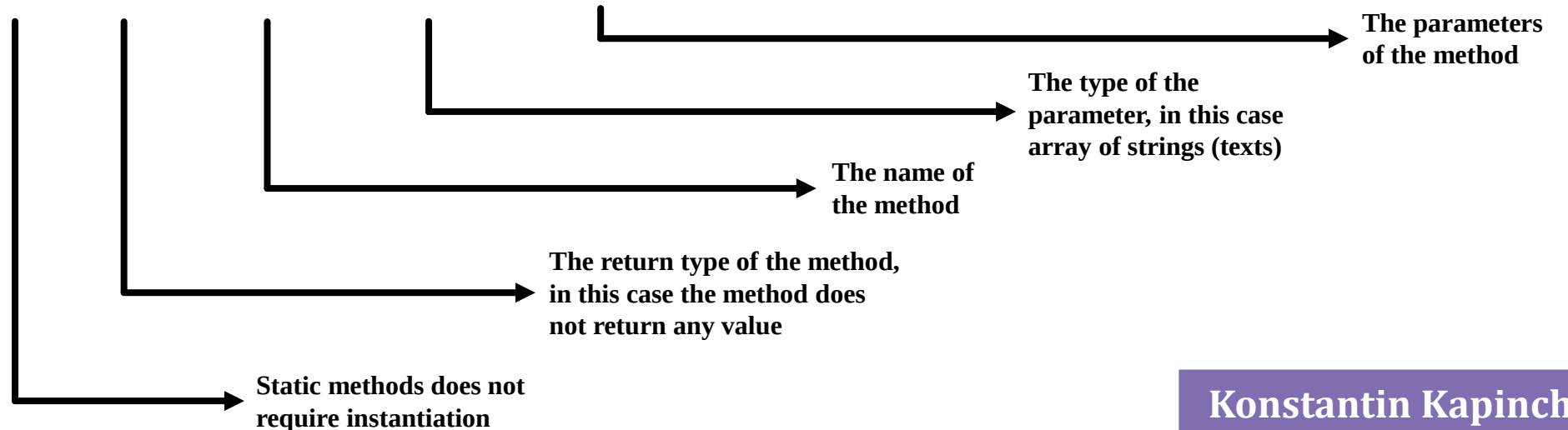
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Syntax of the Language *continued*

- Entry Point

```
static void Main(string[ ] args)
```



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Syntax of the Language *continued*

- Entry Point
 - Method Main will be called first
 - Method PrintHello will be called second

```
namespace MyNameSpace
{
    class HelloWorld
    {
        static void PrintHello()
        {
            System.Console.WriteLine("Hello");
        }
        static void Main(string[] args)
        {
            System.Console.WriteLine("Hello World");
            PrintHello();
        }
    }
}
```

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Syntax of the Language *continued*

- **Methods**
 - A method is a collection of statements
 - The statements are organised as a single block
 - Those statements will run only when the method is called
 - A method can be called multiple times
 - Methods can have input and output
 - Correspond to functions in Mathematics: $y = f(x)$
 - A general term for method is a subroutine¹

¹Subroutines are called functions in C/C++ and Python

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Syntax of the Language *continued*

- **Methods**
 - **Example: Square root function**

```
using System;
namespace MyNameSpace
{
    class HelloWorld
    {
        static double SquareRoot(double r)
        {
            return(Math.Sqrt(r));
        }
        static void Main(string[] args)
        {
            System.Console.WriteLine("Hello World");
            System.Console.WriteLine(SquareRoot(49));
        }
    }
}
```

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The IDE

- **The Integrated Development Environment (IED) is a key tool of the application development process**
- **IDE provides extensive functionality, including:**
 - **Source code editor¹**
 - **Translator²**
 - **Debugger**
 - **Solution/Project Explorer**
 - **And others**

¹Including syntax highlighting

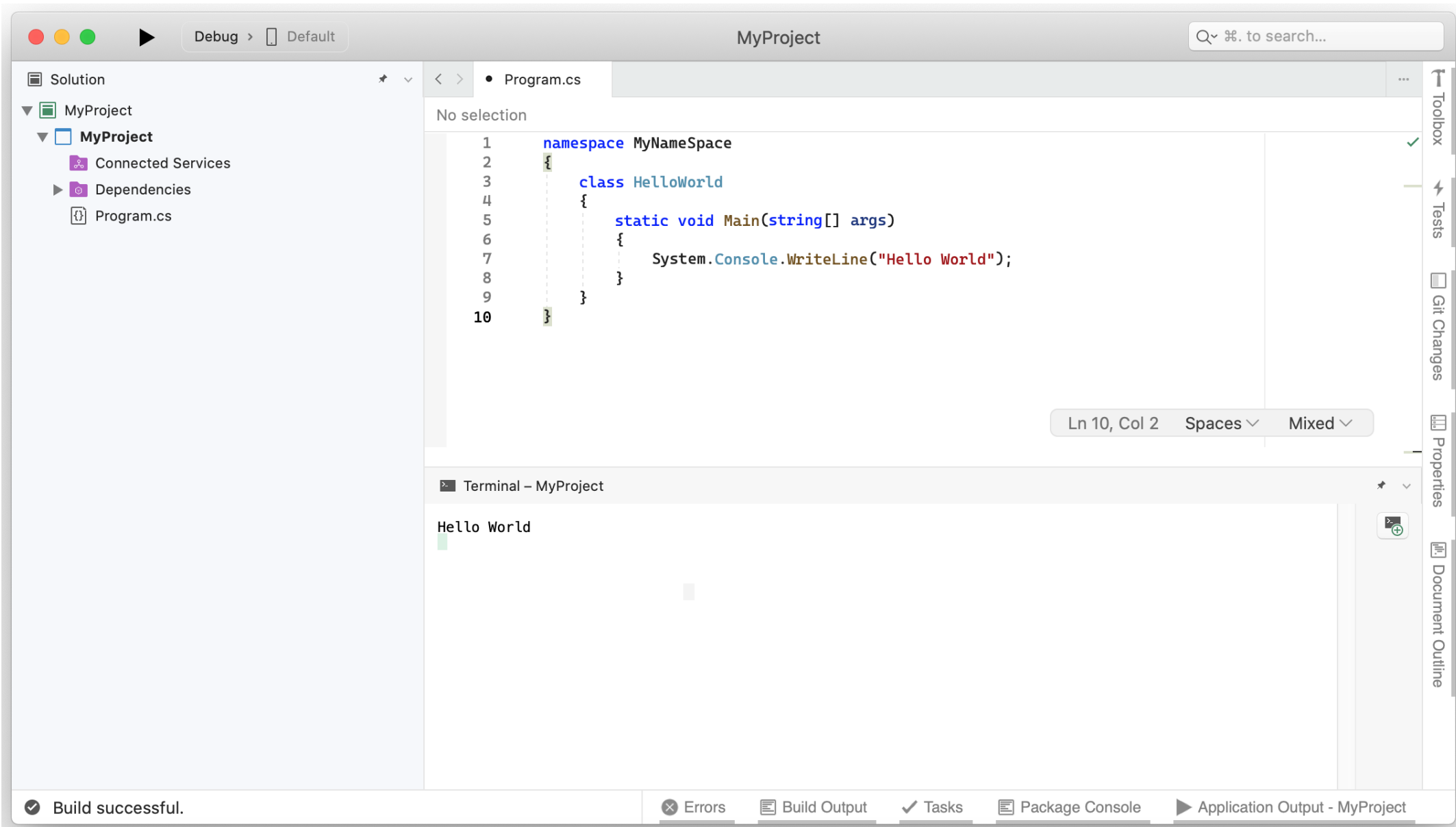
²Compiler, interpreter or both

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The IDE *continued*

- **Microsoft Visual Studio is the IDE of choice for this module**
- **Introduced by Microsoft in 1997**
- **Supports a number of languages, including:**
 - **C++**
 - **C#**
 - **Visual Basic**



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Thank you for your attention.

Any questions?